Neural Vocoders and Neural Codecs as Audio Compressors for Human Speech: From Sequence-based models to GANs

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***Abstract*—This paper, Neural Vocoders and Neural Codecs as Audio Compressors for Human Speech: From Sequence-based Models to GANs, is a high-level minimally technical review of the recent advances in the realm of arbitrary audio generation systems, more specifically, the types of neural vocoders and other neural codecs used in such systems, in the space of machine learning. We will explore the underlying mechanisms allowing for such high-fidelity few-shot outputs, as well as look into the potential implications of such a system if used by wrongdoers in acts of malfeasance. We will also explore solutions to counteract any such malfeasant acts in order to maximally reap the benefits of this new technology while simultaneously minimizing damage.**

***Index Terms*—TTS, Neural Vocoder, Neural Codec, DDPM, Encodec, GAN**

I. Introduction

The past decade has seen immense gains, as well as dramatic breakthroughs in the space of arbitrary audio synthesis conditioned on text-based inputs, ultimately leading to breakthroughs in audio codecs and compression in general. Machine Learning models as data compressors is not a new idea, going back to the work of Shannon. In this paper, we investigate a new class of machine learning models, namely, neural audio codecs. These Neural codecs can be thought of as trained models, that - similar to a generic classical audio codec, as lookup page of some sort, for decoding intermediate audio representations (mel spectrograms, etc) into audio waveforms at a given frequency. More specifically, text-to-speech systems, as well as text-to-music have seen commendable improvements, primarily powered by algorithmic advances in the space of neural networks and end-to-end probabilistic modeling. In this paper, Neural Vocoders and Neural Codecs as Audio Compressors, we provide a survey of the underlying mechanisms of such systems, as well as the likely negative impacts of unabated use of these models, and solutions to minimize such instances. Generally speaking, TTS systems, as of recent times, operate under two fairly distinct paradigms in terms of how they are architected; namely, the cascaded text-to-speech (TTS) systems [Shen et al., 2018, Ren et al., 2019, Li et al., 2019], which primarily leverages pipelines having acoustic models and a vocoder using a mel spectrogram as an intermediate representation to be learned, that would map to an ideal waveform corresponding to the desired output. We also now have text-to-speech (TTS) systems that are based on sequence modeling of discretized audio codec codes based on phenome and acoustic code prompts that map onto the target content and the input speakers' voice [Brown et al., 2020]. The emergence of Denoising Diffusion probabilistic models (DDPMs) has evoked what some might call a renaissance in the space of generative machine learning. Initially, the positive effects of DDPMs have been thought to be constrained to the realm of generative sequence modeling and image generation, but more recent works like Audio-diffusion have shown great improvements in the output fidelity of their DDPM UNET based upsampler and downsampler block [4]. In subsequent sections of this paper, we investigate, on a deeper - more technical level, how these paradigms of text-to-speech (TTS) systems function, as well as Audio-diffusion, and how best to leverage their fairly distinct architectures in order to provide failsafe and contingencies to minimize the likelihood of such systems being used for malfeasant acts.

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II. System overview

TortoiseTTS, from an architectural perspective, can be conceived as five separately trained neural networks that have been pipe-lined together to achieve high-fidelity audio few-shot outputs conditioned on textual input. It includes, as its sub-networks an autoregressive decoder based on GPT(Generative Pre-Trained Transformer), CLVP and CVVP(contrastive voice-voice pre-training) embedding models, a DDPM-based decoder, and, finally, a Univnet-based neural vocoder. These sub-networks in conjunction allow for relatively high-fidelity speech audio outputs based on few-shot inputs.

In multiple ways, the open-source text-to-speech system, AudioDiffusion is similar in architectural design to TortoiseTTS, the primary caveat though is in the diffusion-based upsampler applied to the waveform, as well as the UNet-based vocoder for decoding mel-spectrograms into high-fidelity music output [4].

Now, we arrive at the final sort of text-to-speech architecture - the Large language model(LLM) based systems. These are the latest architectural paradigm being explored as viable methods for high-fidelity speech production given minimal input examples (few/zero-shot systems). The main caveat with such Language-model based text-to-to speech systems is in the intermediate representation used to train the sub-neural networks involved; namely, the use of an audio codec as that intermediate representation [Wang et al., 2020], as opposed to a mel-spectrogram as is the case in cascaded text-to-speech-systems, such as TortoiseTTS.

A. Generic Speech Audio Coding

To provide some context, we will undergo a high-level review of generic speech audio coding mechanisms and standards already in use.

In the field of signal processing, primarily, digital signal processing, a common and somewhat indispensable method is that of signal transformations between different domains (ie, frequency, time, amplitude, etc). Generally speaking, input signals of any form usually exist in a continuous form, meaning that to use such an input signal at any capacity, we would have to initially convert it into a discrete representation, entailing sampling the continuous input waveform using at specific intervals, based on a sampling criterion denoted by 1/Fs, where Fs is the sampling frequency, to return a discretized array of values that coding algorithms can then be applied onto.

Generally, Speech coding involves the application of data compression techniques specifically tailored for digital audio signals carrying speech content, which usually falls between a frequency range of 85 to 255Hz at a sampling frequency of 1600. This process encompasses speech-specific parameter estimation via audio signal processing methods, allowing for the modeling of speech signals. These models, combined with generic data compression algorithms, are employed to encapsulate resulting parameters within a condensed bitstream.

The utility of speech coding extends across various domains such as mobile telephony and Voice over IP (VoIP). In mobile telephony, linear predictive coding (LPC) stands as the predominant technique, while both LPC and modified discrete cosine transform (MDCT) techniques find widespread use in VoIP applications.

The methodologies underpinning speech coding mirror those used in audio data compression and audio coding, leveraging an understanding of psychoacoustics to transmit only data pertinent to the human auditory system. For instance, within voiceband speech coding, solely the information within the frequency band of 400 to 3500 Hz is transmitted, yet the reconstructed signal maintains sufficient intelligibility.

Distinguishing itself from other audio coding forms, speech coding benefits from the simplicity of speech signals and available statistical information regarding speech properties. Consequently, certain auditory information pertinent to general audio coding becomes superfluous within the context of speech coding. Emphasizing the preservation of speech intelligibility and quality, while operating within constrained data transmission, remains a key focus. Moreover, most speech applications necessitate low coding delay to minimize interference with speech interactions.

Given the broad overview of generic speech audio coding, we can proceed to canonical vocoders as a means for encoding/decoding human speech audio. Classical vocoders have seen immense applications in the field of human speech processing. They proceed under two distinct paradigms, namely using a time-based model of the speech waveform such as Linear predictive vocoding, or by employing channel and formant vocoders via cleaving the discretized input signal into frequency components and subsequently modeling those components based on the principles of channel and formant vocoding.

In the space of simulated human speech modeling with the tones and prosody that are closest to the ground truth input, these methods become generally limited by the exponential growth of input values between consecutive transformations and end up not deviating substantially from the ground truth. This is where newer, machine-learning-based neural vocoders and neural codecs become indispensable.

B. Neural Vocoders

Neural vocoders, at their core, constitute trained neural networks that serve as linchpins within the landscape of speech synthesis workflows. Their fundamental role involves translating spectral representations of audio signals into coherent waveforms, a process pivotal in capturing the essence of essential audio attributes effectively. These vocoders function as integral components within this framework, employing sophisticated neural networks to convert spectral information—such as Mel-Spectrograms—into rich, natural-sounding speech waveforms, thus significantly elevating the overall quality and fidelity of synthesized speech.

Specifically engineered to derive waveforms from frequency-based acoustic characteristics like MelSpectrograms or F0 frequencies, vocoders have traditionally relied on DSP-based methodologies, reigning as the leading-edge techniques for vocoder evolution over an extensive period. Despite their efficiency in prompt speech generation, these methods grappled with limitations in the quality of synthesized waveforms, owing to their dependence on specific assumptions and heuristic processes.

However, recent times have witnessed a transformative shift, primarily catalyzed by the ascent of more sophisticated vocoders harnessing the potential of deep neural networks. These innovative approaches aim to substantially enhance the speech quality and naturalness of generated outputs. Diverse neural network-based techniques, spanning from autoregressive methods to Generative Adversarial Networks (GANs) and diffusion-based strategies, have surged in prominence within this domain. They offer promising avenues for augmenting the fidelity and prosody of synthesized speech, leveraging intricate neural network architectures and cutting-edge algorithms to achieve these advancements.

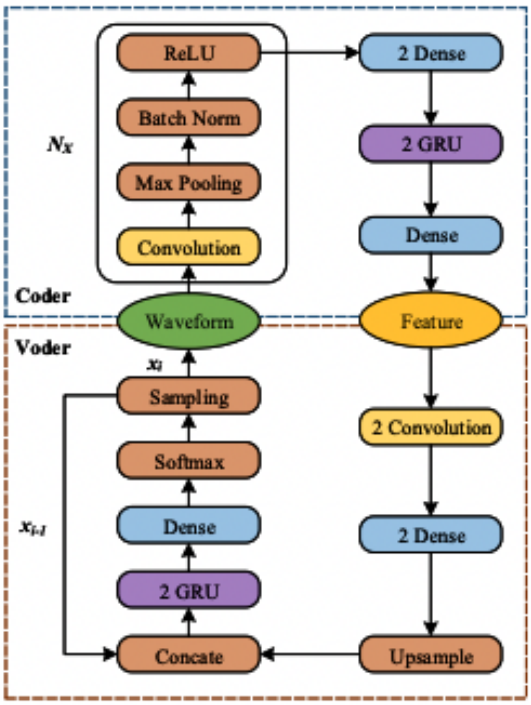

Fig. 1. An overview of a canonical neural vocoder architecture. The model architecture of RawNet mainly consists of two parts: coder and voder. The upper part is the coder network which extracts acoustic features from raw waveform, and the bottom part is the voder network which generates speech from learned features. Adapted from [8]

In forthcoming sections, we will delve into comprehensive explorations, employing real products as case studies, to scrutinize various neural vocoders and their underlying neural network frameworks, encompassing GANs, autoregressive language models, and more. These investigations aim to illuminate the nuanced workings and transformative potential of these neural vocoder/codecs in shaping the landscape of speech synthesis technology.

C. Neural codec

Another fascinating aspect in the realm of neural network-driven human speech audio codecs is the advent of neural codecs grounded in language modeling. These neural codecs, much like neural vocoders, operate as trained neural network models. Their distinctive feature lies in their dependence on the segmentation of the discrete input waveform into tokens, followed by the application of generalized language modeling techniques on these tokenized segments.

These neural codecs represent a burgeoning field, employing sophisticated strategies akin to those utilized in language processing tasks. By harnessing the power of language models, these codecs engage in a process that involves parsing and analyzing the discretized input waveform at a granular token level, enabling them to comprehend and reconstruct speech audio more effectively. These models leverage their understanding of linguistic patterns and contextual information to encode and decode speech signals, promising advancements in speech compression and reconstruction techniques.

In the upcoming segments, we utilize tangible examples - Encodec, and VALL-E[1], as focal points for our investigation into the practical application of neural-codec-based models. This exploration delves into the comparative analysis between these models and neural vocoder-based counterparts concerning two crucial aspects: audio compression efficiency and the nuanced art of achieving high-fidelity human speech synthesis. Our scrutiny involves an in-depth examination of how these models perform in real-world scenarios, shedding light on their respective strengths and limitations in addressing these pivotal tasks within the realm of audio processing and synthesis.

D. Neural Vocoder based systems: Cascaded text-to-speech (TTS) systems for zero-shot TTS

TortoiseTTS has as one of its underlying network modules, a decoder-only generative pre-trained transformer (GPT) network. This submodule is used primarily as an autoregressive decoder and is the crux of much of the functions of the system. The system is based on a generative model that uses autoregressive decoding to produce highly compressed audio data from text inputs and reference clips. The components of the Tortoise system work together to generate natural-sounding, high-fidelity speech outputs that most closely match the input text and reference audio it's being conditioned on.

The autoregressive decoder is the core component of the Tortoise system. It takes in text inputs and reference clips and generates latents and corresponding token codes that represent highly-compressed audio data. The latents are then used by the diffusion decoder to produce a MEL spectrogram that represents the speech output.

To generate natural-sounding speech, the Tortoise system uses a nucleus sampling decoding strategy. This means that the system generates multiple "candidate" latents for each input text and reference clip. The system then uses the CLVP and CVVP models to select the best candidate.

The CLVP model produces a similarity score between the input text and each candidate code sequence. The CVVP model produces a similarity score between the reference clips and each candidate. The two similarity scores are then combined with a weighting provided by the Tortoise user. This allows the system to choose the candidate with the highest total similarity to proceed to the next step.

Once the candidate has been selected, the diffusion decoder consumes the autoregressive latents and the reference clips to produce a MEL spectrogram representing the speech output. Finally, a Univnet vocoder is used to transform the MEL spectrogram into actual waveform data that can be played back as speech.

Overall, the Tortoise system is a highly sophisticated text-to-speech system that uses advanced machine learning techniques to produce natural-sounding, high-fidelity speech output that would most closely match the input text and reference audio. The system's autoregressive decoding, nucleus sampling, CLVP, CVVP, and diffusion decoder components all work

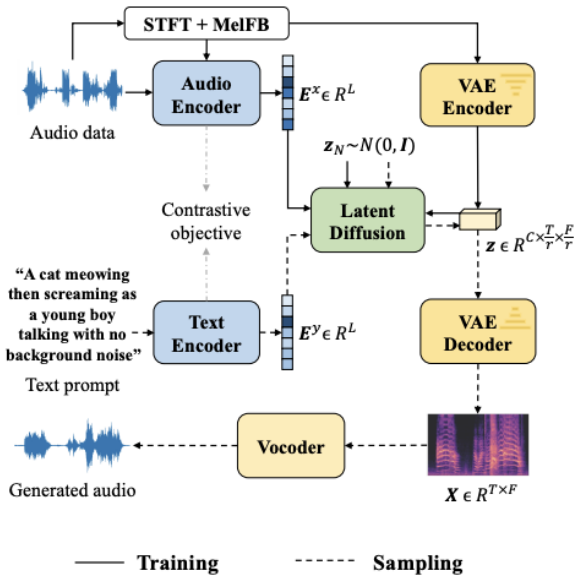

Fig. 2. An Overview of AudioLDM design for the text-to-audio generation. During training, latent diffusion models (LDMs) are conditioned on audio embedding and trained in a continuous space learned by VAE. The sampling process uses text embedding as the condition. Adapted from [2]

together seamlessly as subnetworks or cascaded networks to create a highly effective and efficient speech synthesis pipeline.

E. Cascaded text-to-music(TTM) systems for zero-shot TTA

This section focuses on the underlying mechanism of Audio Diffusion based systems. Text-to-audio (TTA) systems are a technology that has recently gained attention due to their seemingly remarkable ability to synthesize high-fidelity general audio output based on text descriptions as input. Although, it must be noted that, prior studies in the space of TTA have reported high-frequency instances of limited generation quality despite the high computational costs required. This is where the novel text-to-audio (TTA) system named AudioLDM comes in.

AudioLDM is a more recent - novel text-to-audio (TTA) system built on a latent space, which means that it learns continuous audio representations from contrastive language-audio pretraining (CLAP) latents. By using pre-trained CLAP models, we can train LDMs (latent discriminative models) with audio embedding and text embedding as a condition during sampling. This allows AudioLDM to learn the latent representations of audio signals and their compositions without modeling explicitly the cross-modal relationship, which makes it advantageous in both generation quality and computational efficiency.

In addition, AudioLDM is trained on AudioCaps with a single GPU, yet it achieves state-of-the-art TTA performance measured by both objective and subjective metrics such as the Frechet distance. What's more, AudioLDM is the first TTA system that enables various text-guided audio manipulations, such as style transfer, in a zero-shot fashion.

Overall, the proposed AudioLDM system presents a significant improvement over previous TTA systems, as it achieves both high-fidelity generation quality and computational efficiency. Furthermore, its ability to perform text-guided audio manipulations in a zero-shot fashion makes it a highly versatile tool that could be used in various applications, such as virtual assistants, audiobook production, and speech therapy.

F. Neural codec-based systems: EnCodec

EnCodec is a high-fidelity neural audio codec. Similar to other neural codecs, it is a neural network trained specifically to compress audio waveform inputs of any format, including human speech, music, etc, and successively reconstruct the ground truth while preserving a high amount of fidelity.

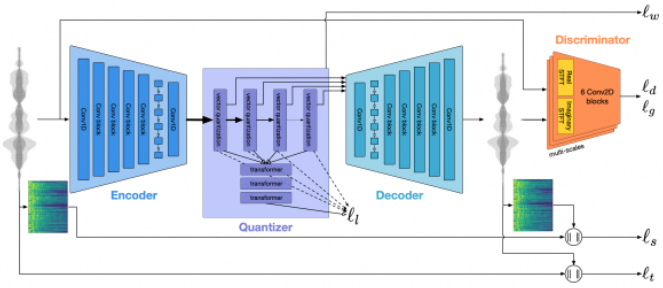

Fig. 3. Encodec: An architectural overview. Adapted from [9]

It makes use of a low-bitrate discretization of an input waveform via an autoencoder-based encoding block with a residual vector quantization bottleneck that produces several parallel streams of audio tokens with a fixed vocabulary. The different streams capture different levels of information of the audio waveform, allowing reconstruction of the audio with high fidelity from all the streams.

EnCodec also breaks ground as being the first neural audio codec, specifically for stereophonic audio at 48kHz[8]. The 48kHz model works at bitrates of 3, 6, 12 and 24kbps[8]. EnCodec also functions at 24KHz with bitrates of 1.5,3,6,12 or 24kbps[8].

These represent a breakthrough in the space of neural audio codecs.

G. Neural codec based systems: LLM based text-to-speech

For this section on language model-based zero-shot text-to-speech system, we will use as our case of study, VALL-E, a zero-shot high-fidelity text-to-speech system that is predicated on using audio codec tokens for intermediate representations, as opposed to mel-spectrograms in cascaded architectures. We can properly conceive VALL-E, then as a neural codec language model such that the neural network tokenizes the input speech and proceeds to use intermediary networks to use those output tokens to build waveforms that correspond to the voice of the speaker, including keeping the speaker's timbre and emotional tone.

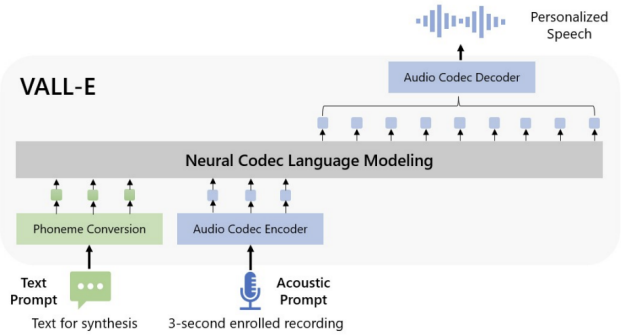

Fig. 4. The overview of VALL-E. Unlike the previous pipeline (e.g., phoneme → mel-spectrogram → waveform), the pipeline of VALL-E is phoneme → discrete code → waveform. VALL-E generates the discrete audio codec codes based on phoneme and acoustic code prompts, corresponding to the target content and the speaker's voice. VALL-E directly enables various speech synthesis applications, such as zero-shot TTS, speech editing, and content creation combined with other generative AI models like GPT-3. Adapted from [1]

H. Comparison between Cascaded text-to-speech systems and language model text-to-speech systems

At the time of writing, neural-codec based language model text-to-speech systems seem to surpass all other text-to-speech systems on the general benchmarks corpus' such as LibriSpeech and VCTK.

	Current Systems	**VALL-E**
Intermediate representation	mel spectrogram	audio codec code
Objective function	continuous signal regression	language model
Training data	$\leq$ 600 hours	60K hours
In-context learning	✗	✓

Fig. 5. A comparison between VALL-E and current cascaded TTS systems. Adapted from [1]

III. Ethical Implications Overview

Such technologies as the ones explicated above pose a latent risk given the ethical implication accompanying their proliferation and unregulated use by both malicious individuals, as well as unethical corporations or autocratic governments. The negative ethical implications of the proliferation of audio-generative models could metastasize as - corporate espionage, use by autocratic governments to exert control, cases of fraud, and as a means to incite violence of any form. The following sections will be a more in-depth survey of each of these ethical implications, as well as possible solutions to curb any such negative outcomes that could occur as a result of proliferation and unregulated access to such powerful systems.

A. Corporate Espionage

As an initial case study for the potential negative use cases of such audio-generative systems, corporate espionage can be conceived as a low-hanging fruit with respect to the myriad of potential malfeasant acts that can be carried out in a scenario of unfettered access to such systems.

To paint an exemplifying illustration, take, for example, the scenario of where a malfeasant actor(s), either within the corporation or an external actor(s), could employ such audio-generative systems to create convincing, high-fidelity audio recordings of both confidential conversations and/or meetings held within the said corporation, which would ultimately be used for nefarious purposes.

This could ultimately include acts such as leaking sensitive information, blackmailing employees, or even as a means to manipulate stock prices. Furthermore, audio-generative systems could also be used to create fake audio evidence, in an attempt to frame certain individuals for supposed illegal activities or to impersonate high-level executives in order to gain access to even more sensitive information.

B. Autocratic Governments

A case for utilizing these new audio-generative systems by autocratic governments, in an effort to more effectively exert control on the populace via propaganda and intimidation. To summarize the primary premise, in the hands of autocratic governments, audio-generative systems can be used to perpetuate propaganda, disinformation campaigns, and psychological warfare, with devastating consequences for their citizenry and beyond.

To explicate the negative effects of autocratic governments utilizing these systems, they can:

- Dissemination of False Information: One of the most obvious negative effects of audio-generative systems being utilized by autocratic governments is the dissemination of false information. These systems can be used to create realistic audio recordings of people saying things they never actually said, or to generate sound effects that make it seem like events occurred that never actually did. This can be used to manipulate public opinion, sow discord, and undermine democratic institutions.
- Amplifying Hate Speech and Incitement to Violence: Autocratic governments can use audio-generative systems to create hateful or inflammatory messages that are designed to incite violence and discord. These messages can be amplified through social media and other digital platforms, leading to an increase in hate crimes and other forms of violence.
- Undermining Free Speech: Audio-generative systems can be used to create deepfake audio recordings that are designed to discredit journalists, activists, or other individuals who speak out against autocratic regimes. These deepfakes can be used to silence dissent and undermine free speech, making it more difficult for people to speak out against injustices.
- Creating Chaos: Audio-generative systems can also be used to create chaos and confusion. For example, autocratic governments could use these systems to create fake emergency broadcasts or sound effects that simulate explosions, gunfire, or other violent events. This could lead to mass panic and disruption, making it easier

for governments to consolidate power and control their populations.

- Eroding Trust: Lastly, the use of audio-generative systems by autocratic governments erodes trust in institutions and democratic processes. When people cannot trust what they see or hear, they are more likely to become disillusioned with democratic institutions and more susceptible to authoritarianism. This could lead to the erosion of democracy and human rights, both within autocratic regimes and in other countries around the world.

So, we see that in the case of autocratic governments, these systems could have very negative impacts on human rights and democratic institutions. As these systems continue to evolve, becoming more sophisticated, we would do well to develop sufficient safeguards to prevent their misuse by such authoritarian regimes.

C. Fraud

As we are aware, audio-generative systems have the perturbing potential to create highly realistic - convincing fake audio recordings of individuals saying things that have not actually been said. Although there are multiple positive applications of this technology, such as in the entertainment industry, or more creatively, to implement high-fidelity virtual assistants. It still poses a major risk for cases of fraud.

Of the more significant negative cases of audio-generative systems being used for malfeasant acts is the case of it being utilized for fraud, which can ultimately cause serious harm to individuals and organizations. Take, for example, a malfeasant actor, in this case, a fraudster, employing the use of audio-generative technology to create fake audio recordings of a CEO or high-ranking executive giving instructions to transfer large sums of money to a given fraudulent account. This would ultimately result in significant financial losses for the organization, as well as reputational damage.

Another negative effect of audio-generative technology being used for fraud is that it can be challenging to detect. Traditional methods of verifying the authenticity of audio recordings, such as voice recognition software or visual cues, may not be effective when dealing with highly sophisticated audio-generative technology. This can make it challenging for individuals and organizations to protect themselves against fraud, as they may not even realize that they are dealing with a fake recording until it is too late.

Finally, the use of audio-generative technology for fraud can undermine public trust in the authenticity of audio recordings and other forms of evidence. If individuals and organizations cannot be confident that the audio recordings they are dealing with are authentic, it can erode trust in the justice system and other important institutions. This could have significant implications for society as a whole, as it could make it more difficult to prosecute criminals and uphold the rule of law.

Overall, the negative effects of audio-generative systems being utilized for cases of fraud are significant and far-reaching. It is essential that individuals and organizations take steps to protect themselves against this risk, such as by implementing robust authentication and verification procedures for audio recordings and other forms of evidence. Additionally, continued research and development in the field of deep fake detection will be crucial to ensuring that this technology is not used for fraudulent purposes.

D. Inciting Violence

The utilization of audio-generative systems in provoking violence presents a host of detrimental impacts on society. One of the most significant repercussions is their potential to propagate hate speech and extremist ideologies. Through the creation of audio content that advocates violence, prejudice, and discrimination against specific groups or individuals, these systems can intensify dangerous concepts, escalating societal tensions.

Moreover, these systems possess the capability to craft synthetic audio that mirrors authentic recordings, blurring the line between fact and fabrication. This blurred authenticity could lead to widespread confusion, misinformation, and a gradual erosion of trust in media and information outlets due to difficulties in discerning genuine content.

The repercussions extend to democratic values, potentially instigating a climate of self-censorship among individuals and groups. Fearing targeting by these systems, there's a risk of withholding expression, thereby stifling diverse perspectives and impeding the open exchange of ideas, which are fundamental to democratic societies.

The use of audio-generative systems in inciting violence may also invoke legal repercussions. In numerous jurisdictions, hate speech and instigation to violence are grave offenses, and those utilizing these systems to disseminate such content could face prosecution and punitive measures.

In sum, the ramifications of deploying audio-generative systems for inciting violence are multifaceted and extensive. To curb their adverse effects, stringent regulation and vigilant oversight are imperative. Stricter controls can help prevent their misuse, ensuring their application solely for ethical and lawful purposes, safeguarding societal well-being and democratic principles.

IV. Risk mitigation systems: Discriminator

Given the multitude of risks linked to the rise of these potent audio-generation techniques, it becomes imperative to establish robust measures for risk mitigation.

A primary avenue to counter the risks inherent in the proliferation of these systems involves the deployment of a discriminator, a pivotal element within the foundational architecture of Generative Adversarial Networks (GANs)[6]. Consider, for instance, the dual components within a canonical GAN: the generator network and the discriminator network. The discriminator, serving as a crucial system within this paradigm, undergoes training as a binary classifier against the generator's outputs.

Implementing a similar discriminatory mechanism could serve as an effective strategy to discern and mitigate potential

risks stemming from the audio waveforms generated by such systems. By training a discriminator network on these synthesized audio waveforms, the effectiveness of such an approach might be limited primarily by the availability of comprehensive datasets and computational resources.

Furthermore, exploring ways to enhance the discriminatory capabilities of such systems could contribute significantly to fortifying the mitigation strategies against risks associated with these audio-generation techniques. Techniques that focus on refining the discriminator's discernment abilities, perhaps through adaptive learning mechanisms or enhanced feature extraction methodologies, could bolster the overall effectiveness of risk management in this domain.

Moreover, establishing robust evaluation frameworks that continuously monitor and assess the discriminatory performance of these systems against generated audio outputs would be paramount. These frameworks could aid in promptly identifying and addressing any vulnerabilities or loopholes that might emerge in the discriminator's efficacy, thereby ensuring a proactive approach to risk mitigation.

In essence, leveraging discriminative mechanisms akin to those found in canonical GAN architectures presents a promising avenue for mitigating risks associated with the proliferation of powerful audio-generation technologies. However, further research and development to optimize these approaches and their implementation remain crucial in bolstering the resilience of such risk mitigation strategies.

V. Conclusion

In summary, the advent of these innovative technologies is poised to trigger a substantial shift akin to what Thomas Kuhn termed a paradigm shift, particularly in coding human speech audio for synthesis purposes. The integration of neural network-based methodologies within vocoders and audio codecs signifies a profound transformation in how we comprehend and manipulate speech-related data. This shift is anticipated to reverberate through various sectors, profoundly influencing social and economic institutions, and necessitating a strategic reorientation across these domains.

Amidst this transformative phase, it becomes imperative to establish fail-safes to mitigate the potential negative repercussions that may arise. Safeguards and preemptive measures should be put in place to minimize adverse outcomes, ensuring that the advancements primarily yield positive outcomes for the betterment of humanity and society at large. This necessitates a responsible and ethical approach towards the integration and deployment of these technologies, guided by a commitment to harness their potential for the greater good while conscientiously managing their potential risks and consequences.

As these technologies shape the landscape of human speech audio synthesis, it becomes essential for stakeholders across various sectors to adapt and evolve their practices in tandem with these advancements. Proactive engagement, collaboration, and ethical considerations are pivotal in harnessing the benefits these innovations offer while responsibly navigating the complexities they might introduce. Ultimately, a judicious and balanced approach to implementing these transformative technologies will determine their true impact on humanity's future.

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